

WEEK 3- LAB PROGRAM

1. Prime Palindrome

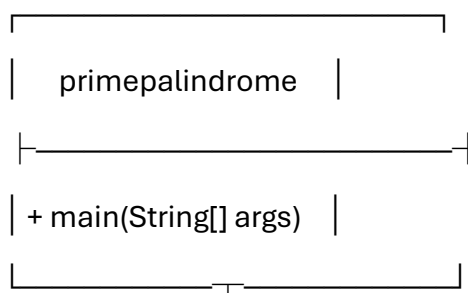
Steps

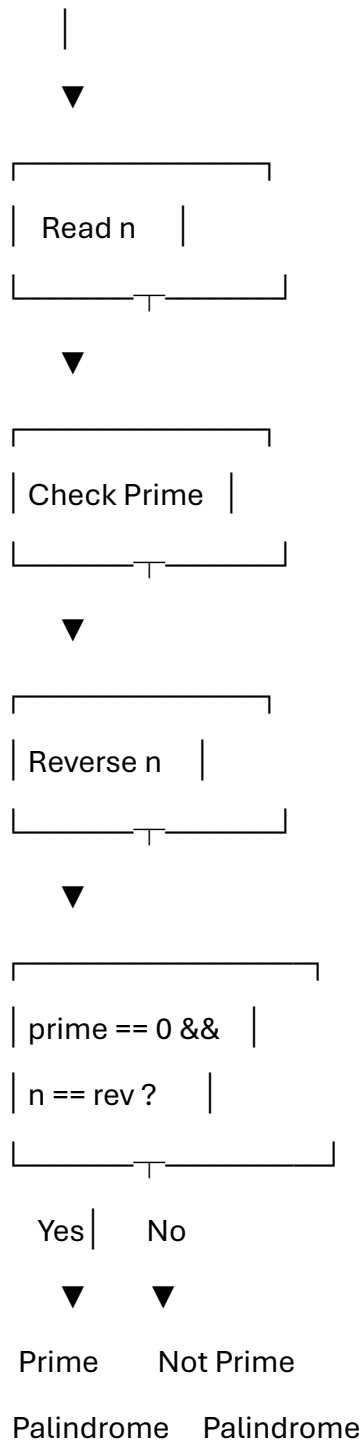
1. Import the Scanner class.
2. Create the primepalindrome class.
3. Read a number from the user.
4. Check whether the number is prime.
5. Reverse the given number.
6. Compare the original number with the reversed number.
7. If the number is both prime and palindrome, display the result.
8. Otherwise, display that it is not a prime palindrome.

Algorithm

1. Start.
2. Read n.
3. Set prime = 0.
4. Check divisibility of n from 2 to n-2.
5. If n is divisible by any number, set prime = 1.
6. Reverse n and store it in rev.
7. If prime == 0 and n == rev, print **Prime Palindrome**.
8. Otherwise, print **Not Prime Palindrome**.
9. Stop.

UML Diagram





Code

```
import java.util.Scanner;

public class primepalindrome {

    public static void main(String[] args) {

        int n,i,prime=0;

        Scanner sc=new Scanner(System.in);
```

```

System.out.println("enter a number");
n=sc.nextInt();
for(i=2;i<n-1;i++){
    if (n%i==0) {
        prime=1;
        break;
    }

}

int temp=n;
int rev=0;
while(temp>0){
    int dig=temp%10;
    rev=rev*10+dig;
    temp=temp/10;
}
if (prime ==0 && n==rev) {
    System.out.println("GIVEN NUMBER IS PRIME PALINDROME");

}
else
    System.out.println("given number is not prime palindrome number");

}
}

```

2. Marks Calculation Using Command Line Arguments

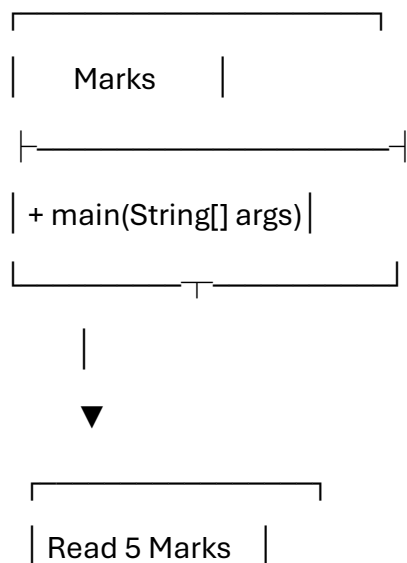
Steps

1. Create a class named Marks.
2. Accept five marks through command line arguments.
3. Convert the arguments from String to int using Integer.parseInt().
4. Calculate the total of five marks.
5. Calculate the average.
6. Display the total and average.

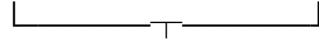
Algorithm

1. Start.
2. Read five marks from args[].
3. Convert each argument into an integer.
4. Calculate:
$$\text{total} = m1 + m2 + m3 + m4 + m5$$
5. Calculate:
$$\text{average} = \text{total} / 5.0$$
6. Display total.
7. Display average.
8. Stop.

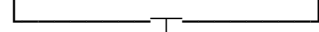
UML Diagram



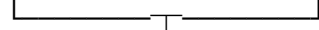
| from args[] |



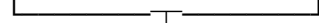
| Convert String |
| to Integer |



| Calculate Total |



| Calculate |
| Average |



| Display Total |
| and Average |



Code

```
class Marks {  
  
    public static void main(String[] args) {  
  
        int m1 = Integer.parseInt(args[0]);  
        int m2 = Integer.parseInt(args[1]);  
        int m3 = Integer.parseInt(args[2]);
```

```
int m4 = Integer.parseInt(args[3]);

int m5 = Integer.parseInt(args[4]);


int total = m1 + m2 + m3 + m4 + m5;


double average = total / 5.0;


System.out.println("Total = " + total);

System.out.println("Average = " + average);

}

}
```

```
PS C:\Users\navvee\OneDrive\Desktop\javaprograms\labprograms\Week 3> java primepalindrome
enter a number
131
GIVEN NUMBER IS PRIME PALINDROME
```

```
PS C:\Users\navvee\OneDrive\Desktop\javaprograms\labprograms\Week 3> java Marks 80
75 90 85 70
Total = 400
Average = 80.0
```